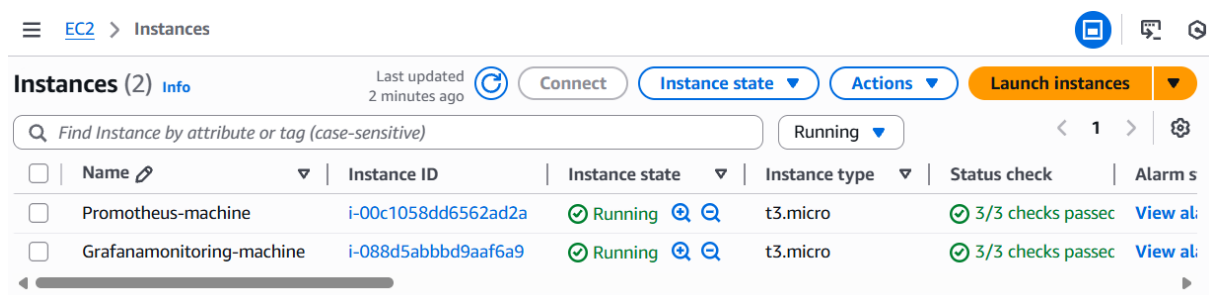


Prometheus & Grafana Assignment

Name: Pooja Prakash Belgaumkar

1. L1 - Create Grafana Dashboard to Monitor CPU and Memory Utilization of Jenkins Build Server using Prometheus Data source

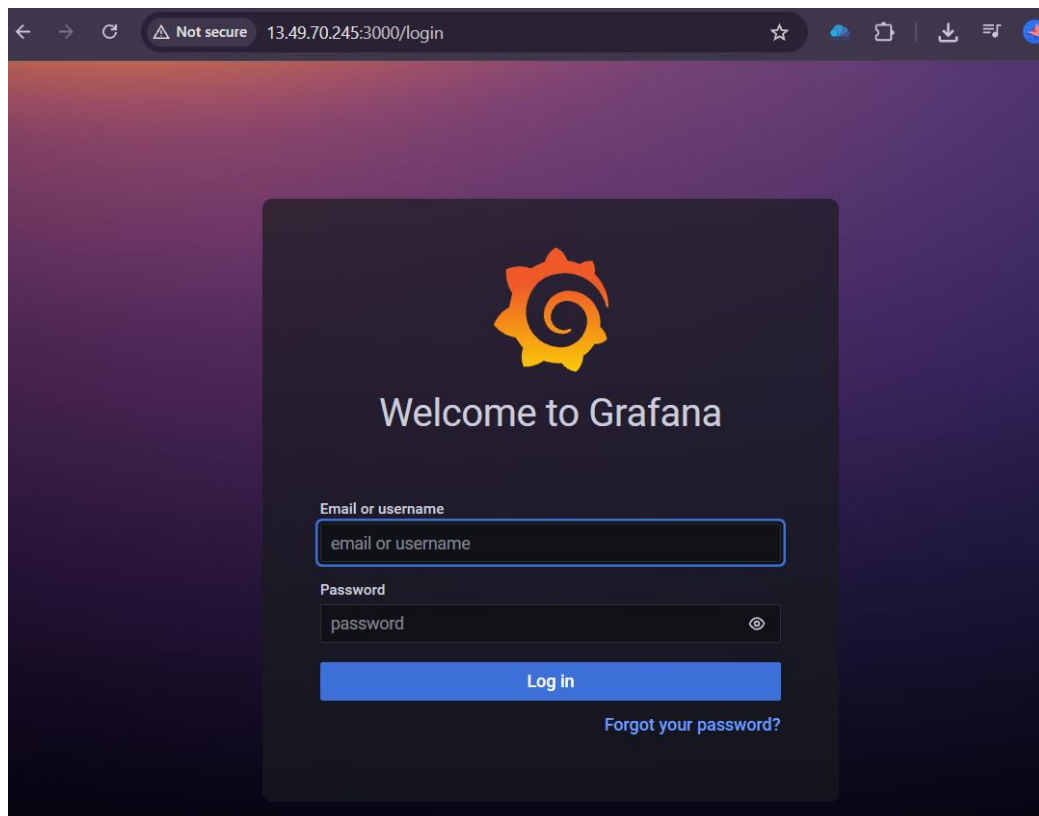
- Create two Ubuntu EC2 machines one for Prometheus and one for Grafana.
- Installation plan:
 - Grafana machine: Install Grafana.
 - Prometheus machine: Install Prometheus, Jenkins and Node Exporter.



The screenshot shows the AWS Management Console 'Instances' page. It displays two EC2 instances in a 'Running' state. The first instance is named 'Prometheus-machine' with ID 'i-00c1058dd6562ad2a'. The second instance is named 'Grafana-monitoring-machine' with ID 'i-088d5abbbd9aaf6a9'. Both instances are of type 't3.micro' and have passed all three status checks. The interface includes a search bar, filters, and a table of instance details.

Name	Instance ID	Instance state	Instance type	Status check	Alarm s
Prometheus-machine	i-00c1058dd6562ad2a	Running	t3.micro	3/3 checks passed	View al
Grafana-monitoring-machine	i-088d5abbbd9aaf6a9	Running	t3.micro	3/3 checks passed	View al

- Grafana-machine setup:
 - Grafana server runs on **port 3000**, so enable this port in the security group.
 - Access the Grafana webpage using the **public IP** of the machine and log in

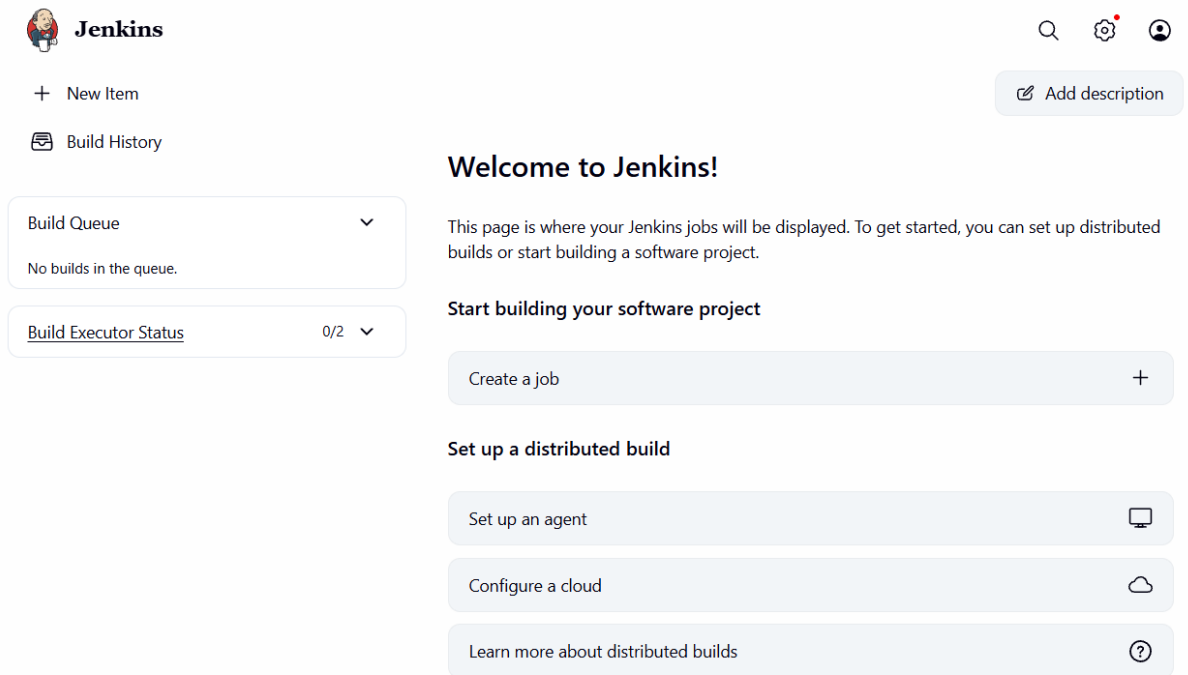


- Prometheus-machine setup:
 - Install Prometheus in the machine

```
root@ip-172-31-0-161:/home/ubuntu# ls
prometheus-2.34.0.linux-amd64  prometheus-2.34.0.linux-amd64.tar.gz
root@ip-172-31-0-161:/home/ubuntu# cd prometheus-2.34.0.linux-amd64
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64#
```

- Installation of jdk and Jenkins and access the Jenkins through 8080 port.

```
openjdk-17-jdk is already the newest version (17.0.18+8-1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 81 not upgraded.
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
root@ip-172-31-13-40:/home/ubuntu#
```



- Install Node Exporter on the Prometheus machine and start it in the background.

```
108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10676343 (10M) [application/octet-stream]
Saving to: 'node_exporter-1.8.2.linux-amd64.tar.gz'

node_exporter-1.8.2.linux 100%[=====>] 10.18M --.-KB/s in 0.04s

2026-02-13 09:30:38 (254 MB/s) - 'node_exporter-1.8.2.linux-amd64.tar.gz' saved [10676343/10676343]

root@ip-172-31-13-40:/home/ubuntu# ls
node_exporter-1.8.2.linux-amd64.tar.gz  prometheus-2.34.0.linux-amd64.tar.gz
prometheus-2.34.0.linux-amd64
root@ip-172-31-13-40:/home/ubuntu# sudo tar xvfz node_exporter-*.linux-amd64.tar.gz
node_exporter-1.8.2.linux-amd64/
node_exporter-1.8.2.linux-amd64/NOTICE
node_exporter-1.8.2.linux-amd64/node_exporter
node_exporter-1.8.2.linux-amd64/LICENSE
root@ip-172-31-13-40:/home/ubuntu# sudo rm node_exporter-1.8.2.linux-amd64.tar.gz
root@ip-172-31-13-40:/home/ubuntu# ls
node_exporter-1.8.2.linux-amd64  prometheus-2.34.0.linux-amd64.tar.gz
prometheus-2.34.0.linux-amd64
root@ip-172-31-13-40:/home/ubuntu# cd node_exporter-1.8.2.linux-amd64
root@ip-172-31-13-40:/home/ubuntu/node_exporter-1.8.2.linux-amd64# sudo ./node_exporter &
[1] 3568
root@ip-172-31-13-40:/home/ubuntu/node_exporter-1.8.2.linux-amd64#
```

- Configure Prometheus by updating the prometheus.yml file to scrape metrics from Node Exporter on port 9100.

```
# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]

  - job_name: "docker"
    static_configs:
      - targets: ["localhost:9323"]

  - job_name: "node_exporter"
    static_configs:
      - targets: ["localhost:9100"]

-- INSERT --
```

35,2

Bot

```
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ls
LICENSE NOTICE  console_libraries  consoles  prometheus  prometheus.yml  promtool
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# vi prometheus.yml
root@ip-172-31-0-161:/home/ubuntu/prometheus-2.34.0.linux-amd64# ./prometheus
ts=2026-02-12T09:49:17.164Z caller=main.go:479 level=info msg="No time or size retention was set so using the default time retention" duration=15d
ts=2026-02-12T09:49:17.164Z caller=main.go:516 level=info msg="Starting Prometheus" version="(version=2.34.0, branch=HEAD, revision=881111fec4332c33094a6fb2680c71fffc427275)"
ts=2026-02-12T09:49:17.164Z caller=main.go:521 level=info build_context="(go=go1.17.8, user=root@12lad7ea5487, date=20220315-15:18:00)"
ts=2026-02-12T09:49:17.164Z caller=main.go:522 level=info host_details="(Linux 6.14.0-1018-aws #18~24.04.1-Ubuntu SMP Mon Nov 24 19:46:27 UTC 2025 x86_64 ip-172-31-0-161 (none))"
ts=2026-02-12T09:49:17.164Z caller=main.go:523 level=info fd_limits="(soft=1024, hard=1048576)"
ts=2026-02-12T09:49:17.164Z caller=main.go:524 level=info vm_limits="(soft=unlimited, hard=unlimited)"
ts=2026-02-12T09:49:17.168Z caller=web.go:540 level=info component=web msg="Start listening for connections" address=0.0.0.0:9090
ts=2026-02-12T09:49:17.168Z caller=main.go:937 level=info msg="Starting TSDB ..."
```

- Access the Prometheus webpage using public IP:9090 and verify that the Node Exporter target is **UP** in the Prometheus UI.

Not secure 51.20.115.124:9090/targets

Prometheus Alerts Graph Status Help

Targets

All Unhealthy Collapse All Filter by endpoint or labels

docker (1/1 up) show less

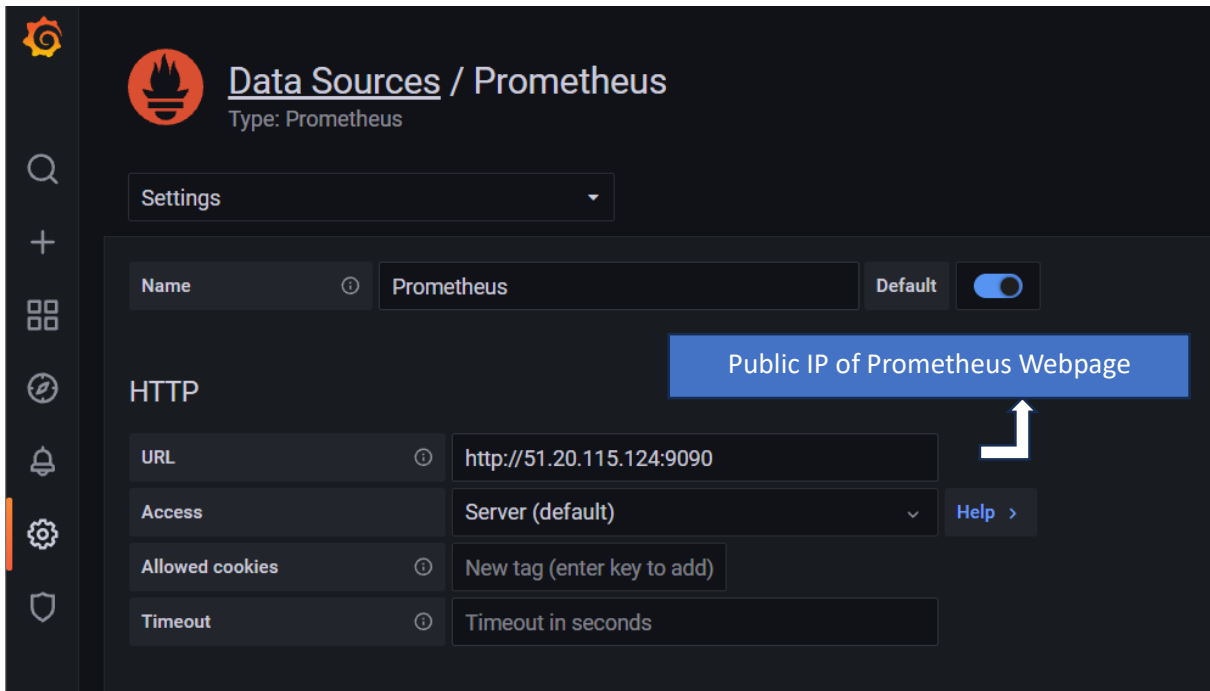
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9323/metrics	UP	instance="localhost:9323" job="docker"	9.846s ago	7.483ms	

node_exporter (1/1 up) show more

prometheus (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	486.000ms ago	6.831ms	

- Connect Grafana to Prometheus by adding Prometheus as a Data Source.



The screenshot shows the Grafana 'Data Sources / Prometheus' configuration page. The interface is dark-themed. On the left is a sidebar with icons for home, search, add, dashboard, alerting, settings, and help. The main content area has a header with the Prometheus logo and title. Below the title is a 'Settings' dropdown. The 'Name' field is 'Prometheus' and the 'Default' toggle is turned on. The 'HTTP' section contains fields for 'URL' (http://51.20.115.124:9090), 'Access' (Server (default)), 'Allowed cookies' (New tag (enter key to add)), and 'Timeout' (Timeout in seconds). A blue callout box with the text 'Public IP of Prometheus Webpage' and a white arrow points to the IP address in the URL field.

Data Sources / Prometheus
Type: Prometheus

Settings

Name Prometheus Default ☒

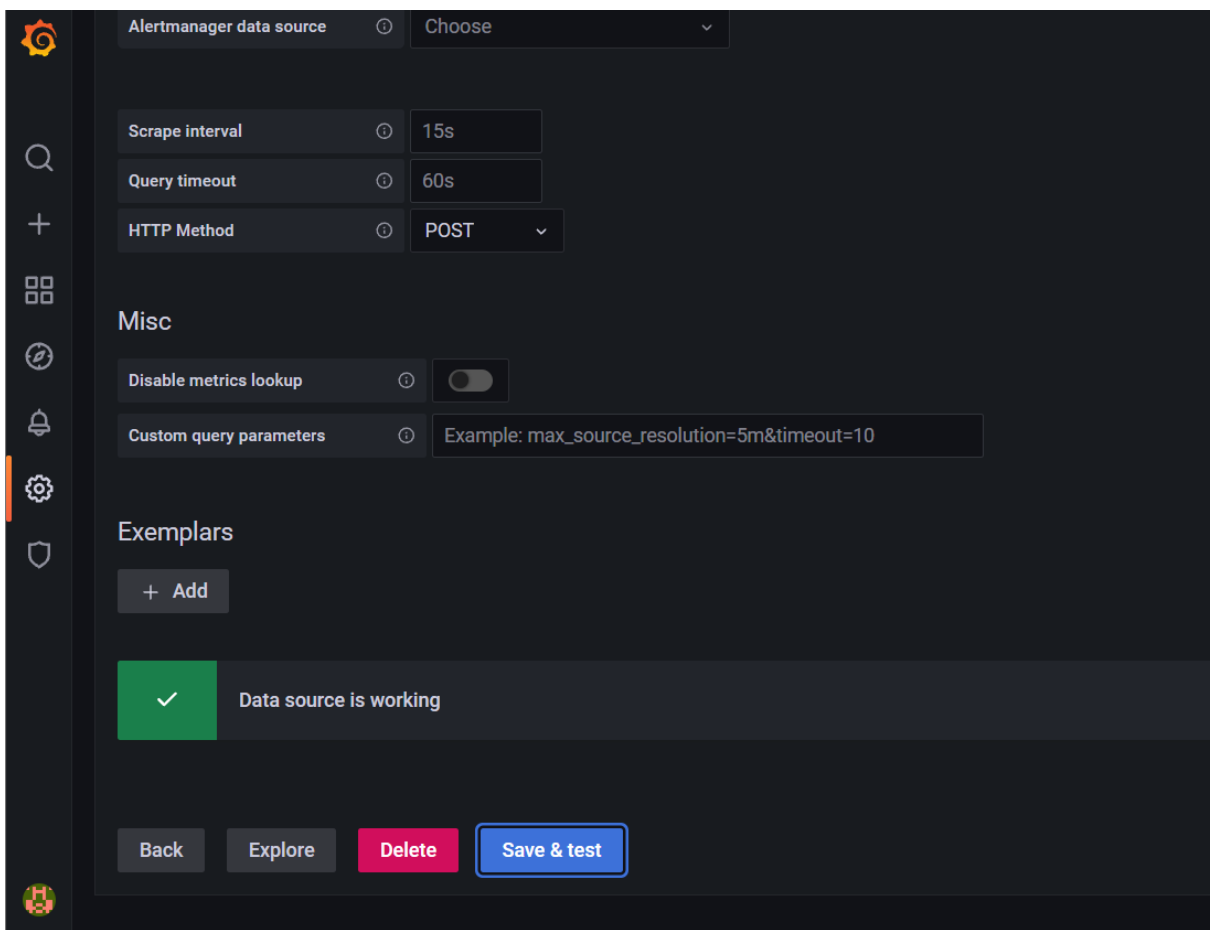
HTTP

URL

Access Server (default) [Help >](#)

Allowed cookies

Timeout



The screenshot shows the 'Alertmanager data source' configuration page in Grafana. The interface is dark-themed. The top section has a dropdown for 'Alertmanager data source' set to 'Choose'. Below this are fields for 'Scrape interval' (15s), 'Query timeout' (60s), and 'HTTP Method' (POST). The 'Misc' section has a 'Disable metrics lookup' toggle (turned off) and a 'Custom query parameters' field with the example 'max_source_resolution=5m&timeout=10'. The 'Exemplars' section has a '+ Add' button. At the bottom, there is a green status bar with a checkmark and the text 'Data source is working'. Below the status bar are buttons for 'Back', 'Explore', 'Delete', and 'Save & test'.

Alertmanager data source Choose

Scrape interval

Query timeout

HTTP Method

Misc

Disable metrics lookup ☐

Custom query parameters

Exemplars

+ Add

✓ Data source is working

Back Explore Delete Save & test

- Add new panel and give metrics to see the graphs of CPU and Memory utilizations.

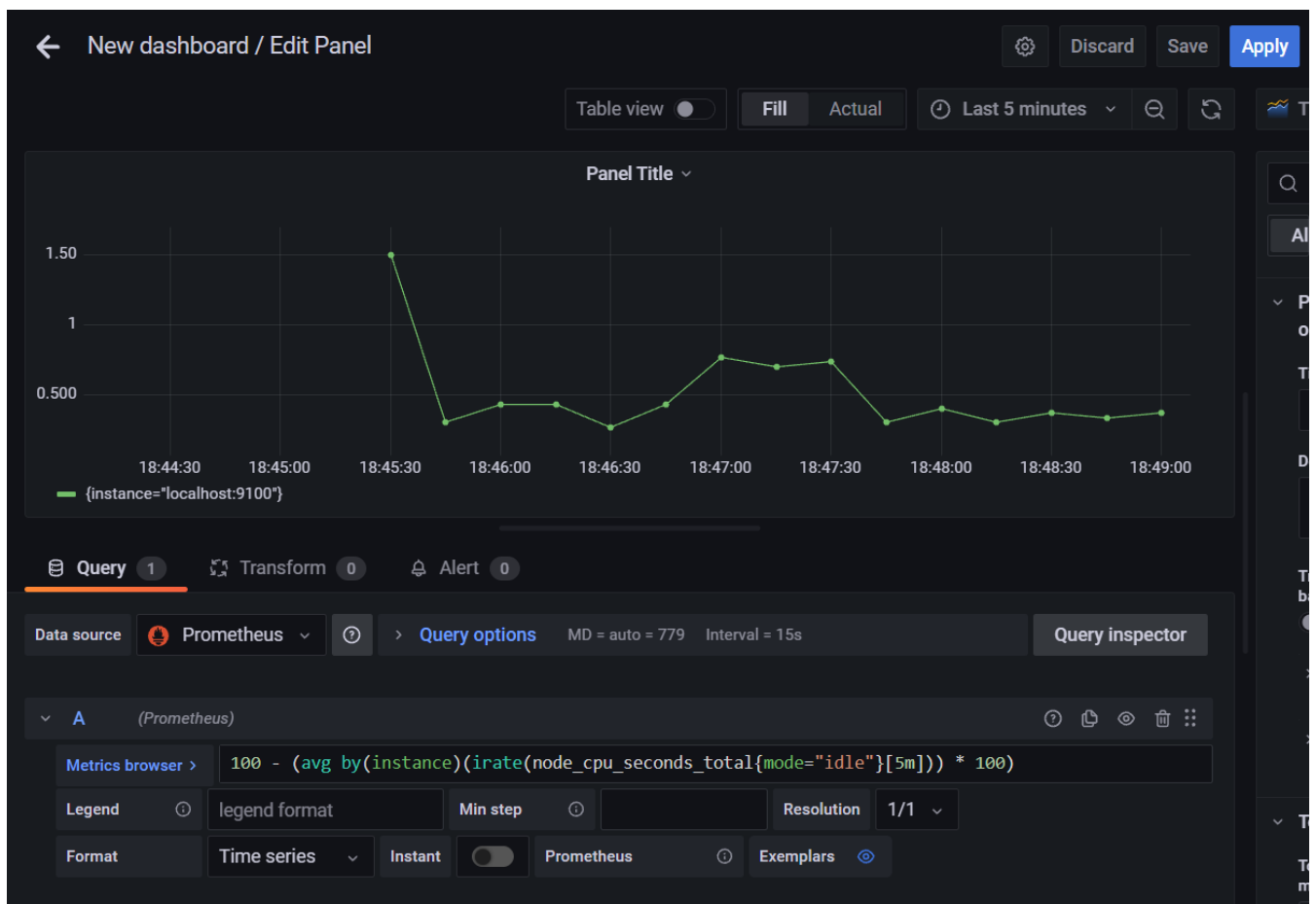
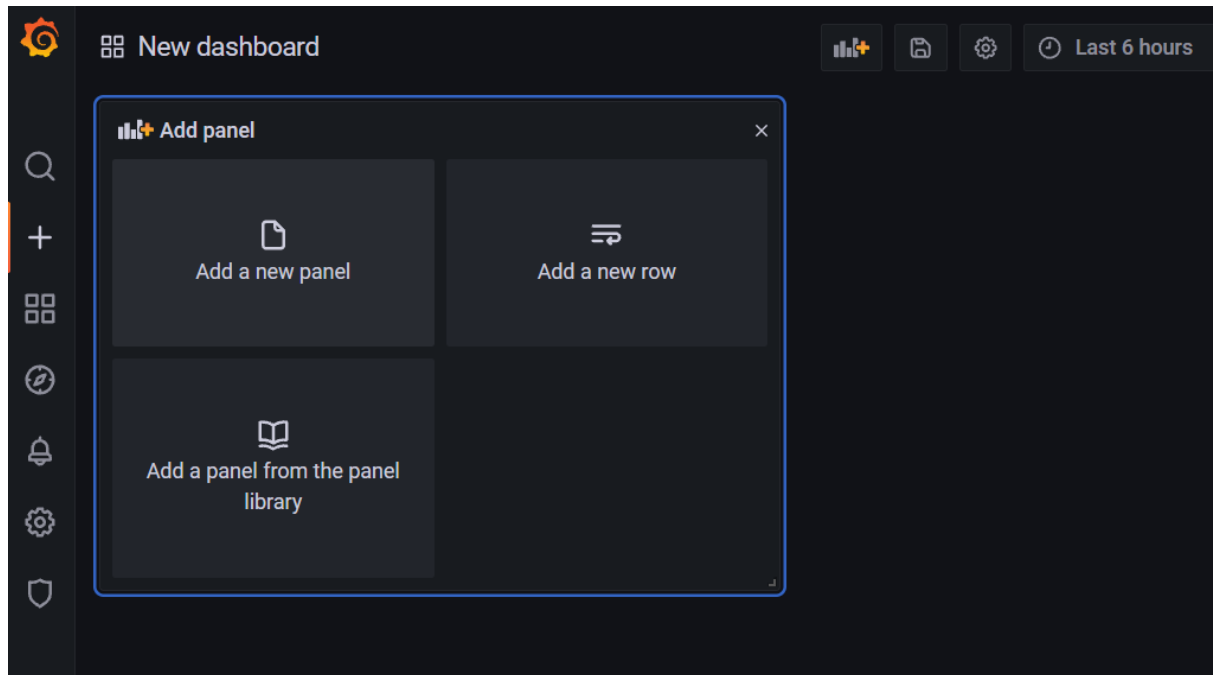
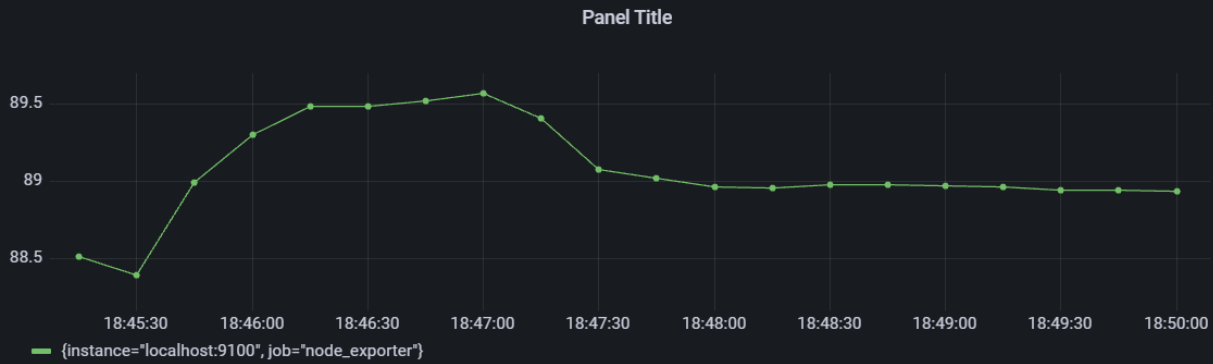


Table view ☐

Fill

Actual

🕒 Last 5 minutes



📄 **Query** 1

🔄 Transform 0

🔔 Alert 0

Data source

Prometheus



> [Query options](#)

MD = auto = 926

Interval = 15s

Query inspector

▼ **A** (Prometheus)



Metrics browser >

`(node_memory_MemTotal_bytes - node_memory_MemAvailable_bytes) / node_memory_MemTotal_bytes * 100`

Legend



legend format

Min step



Resolution

1/1

Format

Time series

Instant



Prometheus



Exemplars

